Storage (With Mod S.O.O. 6247) – General Data

1. General

Fuel storage is accomplished in four fuel tanks, two located beneath the cabin floor and one in each wing leading edge just inboard of the detachable wing tip. The fuselage tanks are fitted longitudinally and referred to as forward and aft tanks. Each fuselage tank consists of four fuel cells, fitted with external and internal lines to provide transfer within each tank. The wing tanks are fitted one in each wing and referred to as left and right wing tanks. The wing tanks, an integral part of each wing, are connected to the engine feed lines from the fuselage tank system. Refer to Figure 1.

A. Fuselage Tanks

There are two fuselage fuel tanks, each consisting of four fuel cells, installed underneath the cabin floor. An external interconnecting manifold, extending between cells 1 and 4 (forward tank) and cells 5 and 8 (aft tank), transfers the fuel within each tank. Contained in each tank are two fuel booster pumps, a level control valve, a fuel low lever float switch, four fuel quantity capacitor units, and a filler neck. Two fuel low pressure switches are mounted externally.

B. Wing Tanks

There are two integral wing fuel tanks, one in each wing, located in the wing leading edge just inboard of the detachable wing tip. Contained in each tank is a level control valve, a filter, a fuel quantity transmitter, a vent pipe, and a filler cap. Mounted outside each tank on the wing outboard nose rib is a fuel pump, a pressure switch, a fuel transfer valve, a refuel shut-off valve, and a filter.

C. Fuel Cells

There is a total of eight fuel cells in the fuel system, all of the flexible rubber type. Each cell contains two molded pipes, on each outboard side, which attach to the vent system lines. With the exception of cells 4 and 5, each cell has a molded fitting which accepts the tank internal interconnecting pipes. In addition, cells 1 and 7 are equipped with filler necks, and the main feed lines to both engines pass through cell 6. Each cell is supported by strips of Velcro tape cemented to the top surface, and along the front and rear faces. On installation, the tape strips engage with matching strips on the underside of the cabin floor, and on the front and rear faces of the compartment frames. An access opening is provided in the bottom of each cell.

D. Fuel Cell Sump Plates

A sump plate is provided to enclose an access opening in the bottom of each fuel cell. The plate contains a row of sealed anchor nuts equally spaced around the machined edge, and has accommodation for various items such as a flap valve, booster pump, fuel capacitor unit, etc. Only the sump plates in cells 2 and 3, and cells 4 and 5, are interchangeable with each other. Flap valves are fitted to the plates used in cells 2 through 6, and 8. The valves in cells 4 and 5 do not have a hole drilled through the center and are only interchangeable with each other.

E. Fuselage Tank Venting

Venting of the fuselage fuel tanks employs two separate and similar vent systems. Each vent system provides a positive tank pressure in flight. A series of interconnected lines from the outboard side of each fuel cell joins a single line which extends up the side of the fuselage and along the inside of the wing to a protruding outlet on the under surface, near the inboard hinge bracket of the outboard fore flap; forward of the vent line outlet is a contoured block which acts as an ice deflector during flight to prevent blockage of the vent. Aircraft with Mod 6/1375 incorporated, have a tunnel over the wing fuel vent outlet to prevent the possibility of vapor ignition due to lightning strikes.

F. Wing Tank Venting

Each wing fuel tank is vented through a single vent tube installed inside the tank. One end of the vent tube is located near the tank roof at the outboard rear of the tank, and the other end terminates flush with the undersurface of the wing at the inboard rear of the tank.

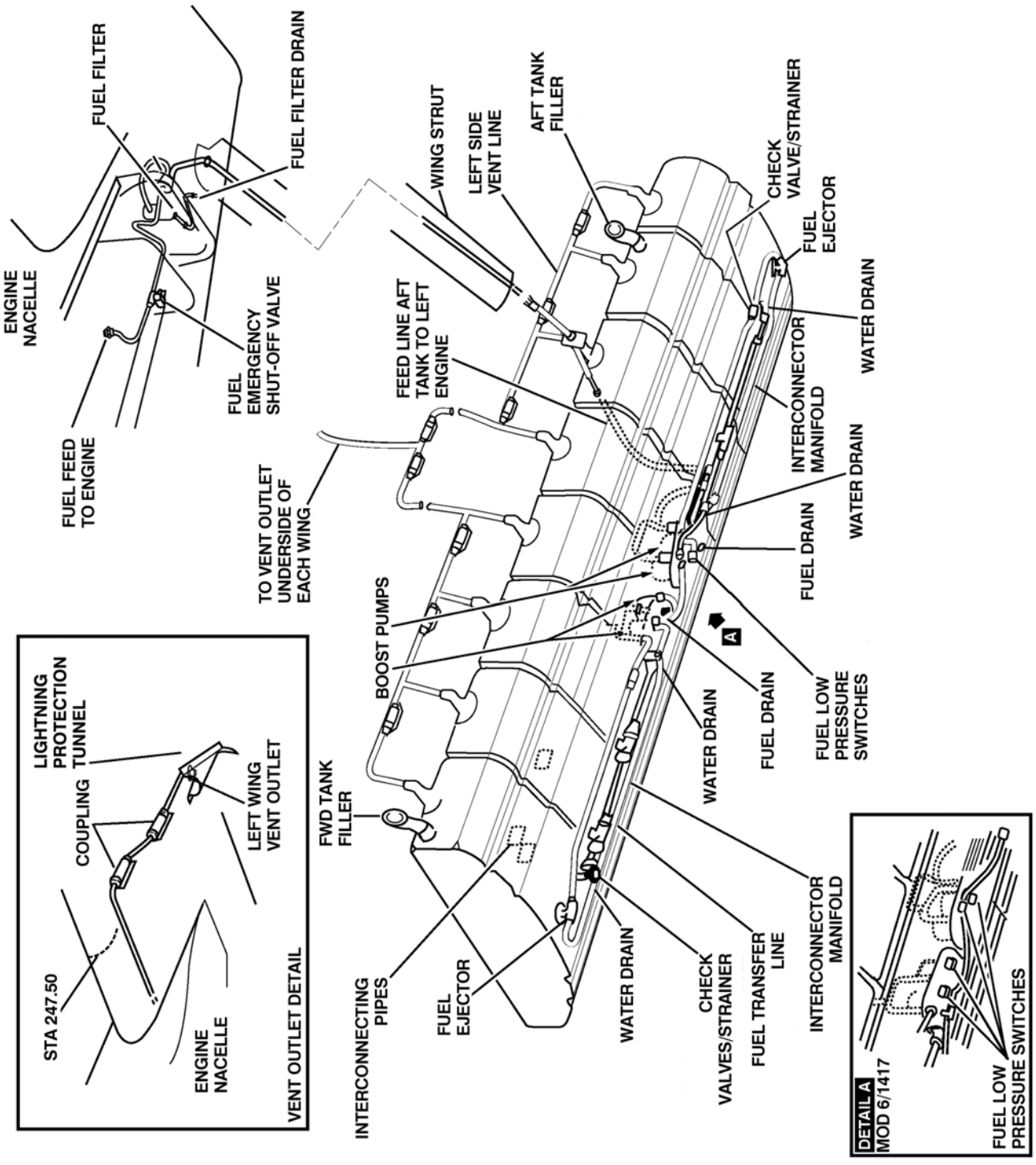
2. Servicing

A. Wing Tank Access Panel – Sealing

1) Check mating surfaces for condition and cleanliness.

2) Check O-ring for condition.

3) Install access panel. Seal gap between access panel edge and wing skin with sealant PR–1776MB (Product Research Co.).



Fuel System Arrangement
Figure 1

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